# ECA5403-ELECTROMAGNETIC FILEDS AND WAVES

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## C02-2

## EXPERIMENT 7

**Aim:**

To determine and plot the capacitance of an isolated sphere and a concentric spherical capacitor for different radii and dielectric constants using MATLAB.

**Test Case (i) – Capacitance of an Isolated Sphere**

Determine the capacitance for a specified radius and relative permittivity.

**MATLAB Program:**

```
clc;
clear;
epsilon0 = 8.854e-12;
a = input('Enter radius of isolated sphere (m): ');
er = input('Enter relative permittivity: ');
epsilon = epsilon0*er;
C = 4*pi*epsilon*a;
disp('Capacitance of isolated sphere (F) =');
disp(C);
```

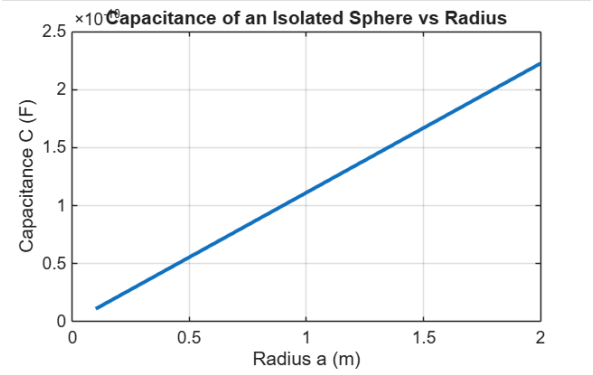
**OUTPUT:**

```
Radius of sphere = 0.50 m
Relative permittivity = 2.00
Capacitance of isolated sphere = 1.112626e-10 F
```

**Test Case (ii) – Variation of Isolated-Sphere Capacitance with Radius**

```
clc;
clear;
epsilon0 = 8.854e-12;
er = 1;
a = linspace(0.1,2,100);
C = 4*pi*epsilon0*er.*a;
plot(a,C,'LineWidth',2); grid on;
xlabel('Radius a (m)'); ylabel('Capacitance C (F)');
title('Capacitance of an Isolated Sphere vs Radius');
```

**OUTPUT:**



**Test Case (iii) – Capacitance of a Concentric Spherical Capacitor**

Determine the capacitance for specified inner radius a, outer radius b and relative permittivity εr.

```
clc;
clear;
epsilon0 = 8.854e-12;
a = input('Enter inner radius a (m): ');
b = input('Enter outer radius b (m): ');
er = input('Enter relative permittivity: ');
epsilon = epsilon0*er;
C = 4*pi*epsilon*a*b/(b-a);
disp('Capacitance of concentric spherical capacitor (F) =');
disp(C);
```

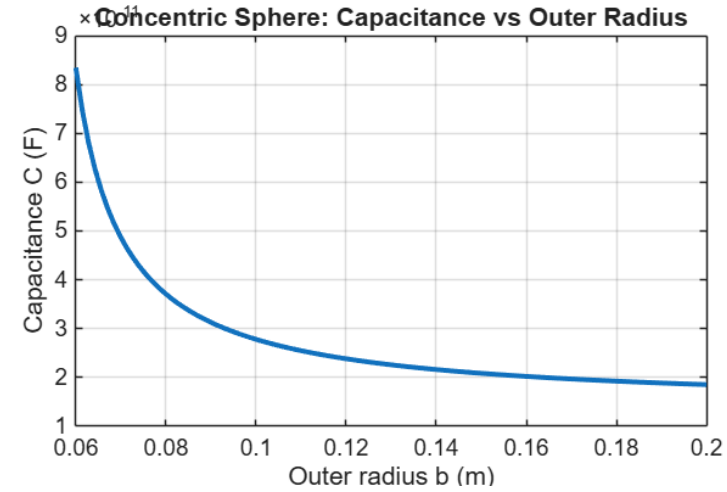
**OUTPUT:**

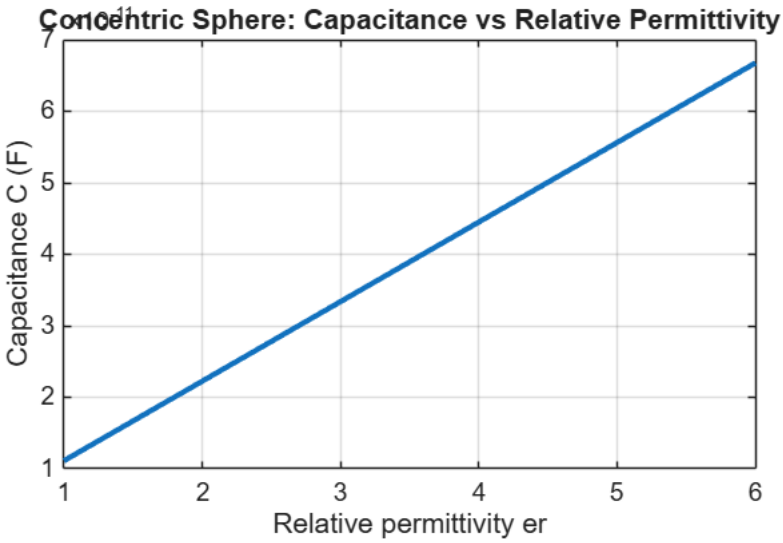
```
Inner radius a = 0.050 m
Outer radius b = 0.100 m
Relative permittivity = 2.00
Capacitance of concentric spherical capacitor = 2.225253e-11 F
```

**Test Case (iv) – Effect of Outer Radius and Relative Permittivity**

```
clc;
 clear;
epsilon0 = 8.854e-12;
a = 0.05;
er = 2.5;
b = linspace(0.06,0.20,100);
C1 = 4*pi*epsilon0*er.*a.*b./(b-a);
figure; plot(b,C1,'LineWidth',2);
grid on;
xlabel('Outer radius b (m)');
 ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Outer Radius');
er = linspace(1,6,100);
b = 0.10;
C2 = 4*pi*epsilon0.*er.*a.*b./(b-a);
figure; plot(er,C2,'LineWidth',2);
grid on;
xlabel('Relative permittivity er');
ylabel('Capacitance C (F)');
title('Concentric Sphere: Capacitance vs Relative Permittivity');
```

OUTPUT:


Concentric Sphere: Capacitance vs Outer Radius
×10^-11
Capacitance C (F)
Outer radius b (m)
9
8
7
6
5
4
3
2
1
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2


Concentric Sphere: Capacitance vs Relative Permittivity
×10^-11
Capacitance C (F)
Relative permittivity er
7
6
5
4
3
2
1
1
2
3
4
5
6

| **Objective** | **Assessment Criterion** | **Excellent – 4** | **Good – 3** | **Satisfactory – 2** | **Needs Improvement – 1** | **Weighting** | **Marks given** | **Marks obtained** |
|---|---|---|---|---|---|---|---|---|
| Define spherical geometry and dielectric | Define spherical geometry and dielectric | Correctly identifies isolated-sphere radius or inner/outer radii and medium permittivity. | Geometry is correct with minor omission. | Some dimensions or material properties are unclear. | Uses invalid geometry or material inputs. | 25 | 2.5 | |
| Implement spherical capacitance equations | Implement spherical capacitance equations | Applies the appropriate isolated/concentric sphere formulas correctly. | Formulas are mostly correct with minor errors. | Partial calculations require guidance. | Uses incorrect equations. | 25 | 2.5 | |
| Analyze radius and permittivity effects | Analyze radius and permittivity effects | Compares cases and validates the expected capacitance trends against theory. | Analysis is mostly correct with minor omissions. | Comparisons or validation are incomplete. | No valid trend analysis is shown. | 30 | 3.0 | |
| Present results and conclusions | Present results and conclusions | Reports geometry, units, calculations and conclusions clearly. | Report is clear with minor gaps. | Report lacks some key information. | Results are unclear or missing. | 20 | 2.0 | |
| TOTAL | | | | | | 100 | 10 | |